Radius & Tangent

Exam code: 9MA0

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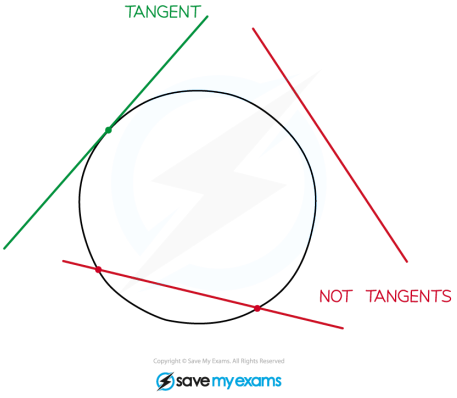
Radius & tangent

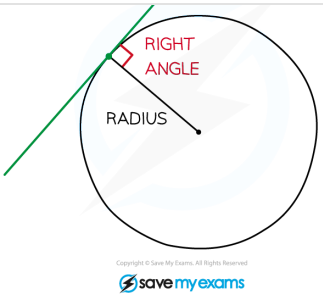
What is the relationship between a tangent and a radius?

- A **tangent** is a line that **touches** a circle at a single point but doesn't cut across the circle

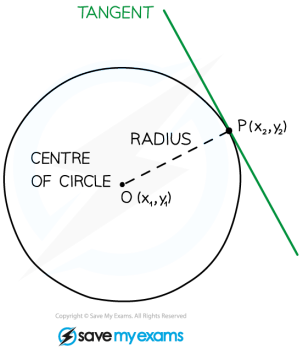


- A tangent to a circle is perpendicular to the radius of the circle at the point of intersection

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



How do I find the equation of a tangent to a circle?



- STEP 1: Find the gradient of the radius ***OP***

$$\text{GRADIENT OF RADIUS} = \frac{y_2 - y_1}{x_2 - x_1}$$

- STEP 2: Find the gradient of the tangent

THE TANGENT IS PERPENDICULAR TO THE RADIUS, SO:

$$\text{GRADIENT OF TANGENT} = -\frac{x_2 - x_1}{y_2 - y_1}$$

- STEP 3: The equation of the tangent is the equation of the line with that gradient that goes through point ***P*** (see Equation of a Straight Line)

Examiner Tips and Tricks

Worked Example

The circle C has equation $(x-5)^2 + (y-2)^2 = 25$.

The point $P(2, 6)$ lies on C.

Find an equation for the tangent to C at point P.

STEP 1: C HAS CENTRE (5,2)

GRADIENT OF RADIUS AT $P = \frac{6-2}{2-5} = -\frac{4}{3}$ ← $\frac{y_2-y_1}{x_2-x_1}$

STEP 2: GRADIENT OF TANGENT AT $P = \frac{3}{4}$ ← $-\frac{x_2-x_1}{y_2-y_1}$

STEP 3: $y = \frac{3}{4}(x-2)+6$ ← LINE THROUGH (2,6) WITH GRADIENT $\frac{3}{4}$

$= \frac{3}{4}x - \frac{3}{2} + 6$

THE EQUATION OF THE TANGENT IS

$y = \frac{3}{4}x + \frac{9}{2}$

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